Lecture 24 (B23): Bayesian Active Learning

Where to sample, BAL algorithm, BAL in economics, BAL + GPs for surrogates

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Motivation: Where to Sample?

Uniform sampling:

- Places points evenly across the domain
- Wastes evaluations in smooth regions
- Cannot adapt to function complexity

Adaptive / active sampling:

- Concentrates points where the function is hard to approximate
- Uses the model's uncertainty to guide selection
- Fewer evaluations for the same accuracy

Key question

Given a budget of N model evaluations, which input points $\{x_1, \ldots, x_N\}$ minimize the overall approximation error?

Bayesian Active Learning: The Idea

Use the GP posterior variance to guide sampling.

Acquisition / score function

$$U(\boldsymbol{x}) = w_{\text{exp}} \cdot \mu(\boldsymbol{x}) + \frac{w_{\text{var}}}{2} \log \sigma^2(\boldsymbol{x})$$

- $\mu(\boldsymbol{x})$: posterior mean (exploitation — sample where function is large)
- $\sigma^2(\boldsymbol{x})$: posterior variance (exploration — sample where uncertainty is high)
- $w_{\text{exp}}, w_{\text{var}} \geq 0$: trade-off weights (distinct from the AR(1) coefficient ρ and the discount factor β used elsewhere).

Pure exploration ($w_{\text{exp}} = 0$): maximize $\sigma^2(\boldsymbol{x})$, i.e., uncertainty sampling.

Mixed ($w_{\text{exp}} > 0, w_{\text{var}} > 0$): balance exploration and exploitation.

For surrogate building, pure exploration (max variance) is often sufficient.

BAL Algorithm

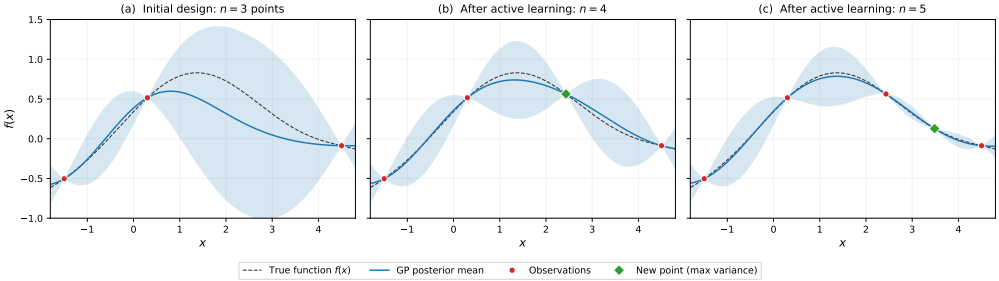
Algorithm 1 Bayesian Active Learning for Surrogates

Require: Initial design $\mathcal{D}_0 = \{(\boldsymbol{x}_i, y_i)\}_{i=1}^{n_0}$, budget N, candidate set $\mathcal{C}$

Fit GP on $\mathcal{D}_0$: compute posterior mean $\mu(\cdot)$ and variance $\sigma^2(\cdot)$
for $t = 1, \ldots, N - n_0$ **do**
    Compute score $U(\boldsymbol{x}) = w_{\mathrm{exp}}\, \mu(\boldsymbol{x}) + \frac{w_{\mathrm{var}}}{2} \log \sigma^2(\boldsymbol{x})$ for all $\boldsymbol{x} \in \mathcal{C}$
    Select $\boldsymbol{x}_{\mathrm{new}} = \arg\max_{\boldsymbol{x} \in \mathcal{C}} U(\boldsymbol{x})$
    Query the expensive model: $y_{\mathrm{new}} = f(\boldsymbol{x}_{\mathrm{new}})$
    Update: $\mathcal{D}_t \leftarrow \mathcal{D}_{t-1} \cup \{(\boldsymbol{x}_{\mathrm{new}}, y_{\mathrm{new}})\}$
    Refit GP on $\mathcal{D}_t$
end for
return Trained GP surrogate

BAL in Action: Progressive Refinement

BAL selects each new point where the posterior variance is largest (green diamond), causing the credible band to shrink exactly where it was widest.



(a) 3 initial observations (red dots); GP posterior (blue) deviates in data-sparse regions with wide 95% CI.

(b) Acquisition function picks max-variance point (green diamond); posterior tightens locally.

(c) Second iteration fills the remaining gap — 5 points suffice to track the true function.

Key contrast with uniform: BAL concentrates points where the function is hardest to approximate.

BAL in Economics

Grid-free approximation

BAL + GP = grid-free adaptive sparse grids:

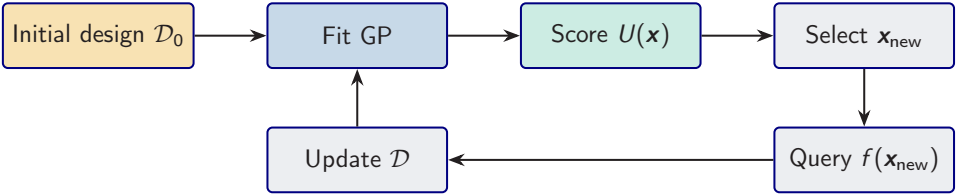
- No structured grid needed
- Refinement via posterior variance
- Scales to $d \leq 10$

Applications

- **Optimal carbon tax**
- **Dynamic incentives** (kinks)
- **Asset pricing** (moderate d)

References: Renner & Scheidegger (2018); Kübler, Scheidegger & Surbek (2025).

Combining BAL and GP for Surrogates



repeat until budget exhausted

Why this combination works

- **GP provides UQ:** posterior variance tells us where the approximation is uncertain.
- **BAL selects training points:** maximize information gain per model evaluation.
- **Surrogate = trained GP:** the same GP used for selection serves as the surrogate.

Ideal for $d \leq 10$ when each model evaluation costs minutes to hours.

Summary: Bayesian Active Learning

1. **BAL** = principled data selection using GP uncertainty.
2. **Score function** balances exploration (high σ^2) and exploitation (high μ).
3. **Faster convergence** than uniform sampling, especially for functions with kinks or local features.
4. **Application:** surrogate construction for moderate-dimensional problems.

Hands-on

Notebook: `02_GP_and_BAL.ipynb` — GP from scratch, scikit-learn GP, and BAL comparison.